

Writing the Equation of a Parabola

Conic Sections Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Write the standard equation of a parabola given its vertex and focus
- Determine the correct standard form based on the direction the parabola opens
- Convert the standard equation of a parabola to its general form

Problems

1. Find the standard equation of the parabola with vertex at the origin and focus at the point below.

$F(1, 0)$

2. Find the standard equation of the parabola with vertex at the origin and focus at the point below.

$F(0, 3)$

3. Find the standard equation of the parabola with vertex at the origin and focus at the point below.

$F(-4, 0)$

4. Find the standard equation of the parabola with vertex at the origin and focus at the point below.

$F(0, -5)$

5. Find the standard equation of the parabola with the vertex and focus given below.

Vertex: $(1, 2)$, Focus: $(4, 2)$

6. Find the standard equation of the parabola with the vertex and focus given below.

Vertex: $(3, -1)$, Focus: $(3, 4)$



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7. Find the standard equation of the parabola with the vertex and focus given below.

Vertex: $(-2, 5)$, Focus: $(-2, 2)$

8. Write both the standard equation and the general equation of the parabola with the vertex and focus given below.

Vertex: $(2, 1)$, Focus: $(2, 4)$

9. Write both the standard equation and the general equation of the parabola with the vertex and focus given below.

Vertex: $(-1, 3)$, Focus: $(2, 3)$

10. A parabola has a vertex at the point below and its focus is 4 units directly to the left of the vertex. Write the standard equation and then convert it to general form.

Vertex: $(5, -2)$



Writing the Equation of a Parabola — Answer Key

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Answer Key

1. Answer: $y^2 = 4x$

- Vertex is at (0,0), focus is at (1,0), so the parabola opens to the right.
- Use the formula $y^2 = 4px$.
- $p = 1$ (distance from vertex to focus).
- Substitute: $y^2 = 4(1)x = 4x$.

2. Answer: $x^2 = 12y$

- Vertex is at (0,0), focus is at (0,3), so the parabola opens upward.
- Use the formula $x^2 = 4py$.
- $p = 3$ (distance from vertex to focus).
- Substitute: $x^2 = 4(3)y = 12y$.

3. Answer: $y^2 = -16x$

- Vertex is at (0,0), focus is at (-4,0), so the parabola opens to the left.
- Use the formula $y^2 = 4px$, where p is negative.
- $p = -4$.
- Substitute: $y^2 = 4(-4)x = -16x$.

4. Answer: $x^2 = -20y$

- Vertex is at (0,0), focus is at (0,-5), so the parabola opens downward.
- Use the formula $x^2 = 4py$, where p is negative.
- $p = -5$.
- Substitute: $x^2 = 4(-5)y = -20y$.

5. Answer: $(y-2)^2 = 12(x-1)$

- The focus and vertex share the same y -coordinate, so the parabola opens to the right.
- Use the formula $(y - k)^2 = 4p(x - h)$.
- $h = 1$, $k = 2$, $p = 4 - 1 = 3$.
- Substitute: $(y - 2)^2 = 4(3)(x - 1) = 12(x - 1)$.

6. Answer: $(x-3)^2 = 20(y+1)$

- The focus and vertex share the same x -coordinate, so the parabola opens upward.
- Use the formula $(x - h)^2 = 4p(y - k)$.
- $h = 3$, $k = -1$, $p = 4 - (-1) = 5$.
- Substitute: $(x - 3)^2 = 4(5)(y + 1) = 20(y + 1)$.

7. Answer: $(x+2)^2 = -12(y-5)$

- The focus and vertex share the same x -coordinate and the focus is below the vertex, so the parabola opens downward.



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- Use the formula $(x - h)^2 = 4p(y - k)$.
- $h = -2$, $k = 5$, $p = 2 - 5 = -3$.
- Substitute: $(x + 2)^2 = 4(-3)(y - 5) = -12(y - 5)$.

8. Answer: Standard: $(x-2)^2 = 12(y-1)$; General: $x^2 - 4x - 12y + 16 = 0$

- Parabola opens upward; use $(x - h)^2 = 4p(y - k)$.
- $h = 2$, $k = 1$, $p = 3$. Standard form: $(x - 2)^2 = 12(y - 1)$.
- Expand: $x^2 - 4x + 4 = 12y - 12$.
- Rearrange to zero: $x^2 - 4x - 12y + 16 = 0$.

9. Answer: Standard: $(y-3)^2 = 12(x+1)$; General: $y^2 - 6y - 12x - 3 = 0$

- Parabola opens to the right; use $(y - k)^2 = 4p(x - h)$.
- $h = -1$, $k = 3$, $p = 2 - (-1) = 3$. Standard form: $(y - 3)^2 = 12(x + 1)$.
- Expand: $y^2 - 6y + 9 = 12x + 12$.
- Rearrange to zero: $y^2 - 6y - 12x - 3 = 0$.

10. Answer: Standard: $(y+2)^2 = -16(x-5)$; General: $y^2 + 4y + 16x - 76 = 0$

- Focus is 4 units to the left, so $p = -4$ and the parabola opens left.
 - Use $(y - k)^2 = 4p(x - h)$ with $h = 5$, $k = -2$, $p = -4$.
 - Standard form: $(y + 2)^2 = -16(x - 5)$.
 - Expand: $y^2 + 4y + 4 = -16x + 80$.
 - Rearrange to zero: $y^2 + 4y + 16x - 76 = 0$.
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